Villarrica Homestead Master Plan v6.1

45 ha estate · household of three · redesigned private compound and house · harvest and nutrient re-run

Private home, Villarrica, Guairá, Paraguay. Replaces v5.2. Every yield, cost and distance in this document is a planning estimate (±20% on food values) and nothing here has been tested or surveyed.

Goal. A calm, private, high-quality daily life (chill, TV and sport, cooking, guitar, gym, swimming, nature, pets) on top of a productive land that feeds three adults almost entirely from the property, with reliability first, then low labour, then efficiency. Life-critical systems (water, power, freezers, animal welfare) win any conflict between comfort and production.

Headline decisions in v6.1

- **45 ha, but the intensive core stays compact.** About 17.9 ha is allocated; about 27 ha stays native forest and wildlife reserve. The mudroom is still within about 150 m of every daily chore except the goat hub (about 175 m).
- **Household of three** (the 125 kg man plus two women). Food need scales ×1.12 on the v5.2 list, then the +30% margin is applied. Annual crops are planted at 1.5× the minimum area because no yield is tested.
- **Walled compound 70 × 72 m (5,040 m²)** with a south gate, an east service gate that keeps harvest and trucks away from the veranda, a guard-dog run that watches the gate, and a 4 m tree belt inside all four walls.
- **House 28 × 10 m (280 m²)**, single storey, three bedrooms, dedicated dark media lounge, bigger kitchen, 98 m² veranda (2.0 m roof plus 1.5 m deciduous pergola so winter sun reaches the floor), five-freezer room.
- **Leisure layer (new in v6.1):** outdoor sound zones, garden cinema, a games pavilion, hot tub, multi-sport court, a stargazing deck, a 2.7 km loop track and ~3 km forest trail, a wetland bird hide, a dam-side shelter with fishing deck, bikes and an offline media server (section 6b).
- **Annex 66 m²** with a 39 m² gym (sized for a heavy lifter) and a 27 m² sealed music room, opening onto a patio at the end of the 25 × 4.5 m lap pool.
- **Power:** 18 kWp solar, 30 kWh usable battery (about 3.8 days of critical loads), 10 kVA generator with ATS. **Water:** 80 m³ rain tanks, 9,000 m³ dam, well; peak demand re-estimated at about 18 m³/day (see section 10).

1. What changed from v5.2, and why

Area	v5.2	v6.1	Why
★ Land	2.0 ha, 1.66 ha active	45 ha; 17.9 ha allocated, ~27 ha native reserve	Land stopped being the constraint; labour, water and capital are. Compact core, generous buffers.
★ People	2 people (man + one woman) in the targets table, mislabelled	3 adults: man 4,400 kcal / 245 g protein; two women 2,200 kcal / 100 g each; total 8,800 kcal / 445 g	The v5.2 label "man + one woman" for 8,800 kcal was wrong; its food list adds up to ~7,880 kcal and ~430 g, i.e. about 90% of the 3-person energy target.
★ Compound	2,058 m² core (~2,400 m² with belts), one gate	5,040 m² walled, two gates, 4 m belt inside the wall	Room for garden, orchard, pool, annex, dogs and privacy without crowding; security line is the wall.
★ House	235 m², 2 bedrooms, 3.0 m veranda	280 m², 3 bedrooms, dark media lounge, bigger kitchen, 3.5 m veranda (98 m²)	Third adult; TV, sport, gaming and cooking get proper rooms; veranda is the main living room.
★ Veranda sun	3.0 m eave "passes winter sun" (unverified)	2.0 m solid roof + 1.5 m deciduous pergola	Geometry check (section 5): at latitude ~25.8°S the June noon sun is ~41° high, so a 3.5 m solid eave at 3.0 m height shades the whole floor in winter. The pergola fixes that.
★ Leisure layer	Pool, fire pit, gym, music, TV wall	Adds outdoor sound, garden cinema, stargazing deck, forest trails, bird hide, dam-side shelter and fishing deck, camp pad, bikes (section 6b)	You asked for everything worth having that improves daily enjoyment on the land.
★ Annex	49.5 m²: gym 24.8 + music 24.8	66 m²: gym 39 + music 27	A 125 kg lifter needs rack, bench, plates, cables and cardio; 24.8 m² is too small.
★ Freezers	4 × 300 L	5 × 300 L (1,500 L)	Cook-week volume rises ~10% (about 630 L per cycle) plus whole tomato, pepper and fruit.
★ Grain	1,800 m² duck maize	0.5 ha duck maize + 0.5 ha rotation (1.0 ha)	Earlier "1.5 ha grain" proposal was oversized. 0.5 ha covers ~1,800 kg duck feed even at single sowing.
★ Water	Peak ~7 m³/day (ponds left TBD)	Peak ~18 m³/day, dam added	v5.2 left pond evaporation as TBD; 2,500 m² of ponds plus a 1.5 ha orchard dominate peak demand.
★ Power	4 freezers, ~11–18 kWh battery	18 kWp, 30 kWh battery, 10 kVA generator	More loads (pool, aerators, AC, cameras, Starlink, gym) and the third adult.

Note: the v5.2 plan quoted peak water demand as ~7 m³/day for the 2 ha version; its ponds line was "TBD", so that figure is not comparable with section 10.

Open assumptions that I could not verify from the documents

- Plot shape (600 × 750 m, rectangular), slope (falls east), prevailing wind (NE summer breeze, cold south wind), and public road on the south boundary are all assumptions. Re-place blocks with the same rules once surveyed.
- Existing vegetation is unknown. If part of the 45 ha is already forest, keep it where it stands and fit the blocks around it.
- Paraguayan forest-reserve and water-body setback rules are not checked here. Confirm with INFONA and MADES before clearing or digging.
- Veranda depth: the plan keeps your 3.5 m. Roof height and pergola design should be checked with a sun-path diagram for the actual survey orientation.

2. Critical numbers (v6.1)

Item	Figure	Notes
★ Land	45 ha (assumed 600 × 750 m)	~17.9 ha allocated to named blocks, ~27 ha native forest, wildlife and spare
★ Household targets	8,800 kcal and 445 g protein per day	Man 4,400 kcal / 245 g (125 kg target, building muscle); each woman 2,200 kcal / 100 g; table error ~20%
★ Compound	70 × 72 m = 5,040 m²; wall 2.4 m high, 284 m long	4 m tree belt inside all walls; 3 gates (main 5 m, pedestrian 1 m, service 3 m)
★ House	280 m² (28.0 × 10.0 m) + 98 m² veranda	3 bedrooms, media lounge, potting room, 5-freezer room, mudroom with inverter wall
★ Annex + pool	Annex 66 m² (+27 m² patio); pool 25 × 4.5 m, 112.5 m², ~150 m³	Pool + deck footprint 210 m²; plant room 20 m²
★ Annual crops	3,365 m² planted (min. 2,248 m²) + 270 m² rain shelters	Planted at 1.5× minimum because yields are untested
★ Orchards	Main 1.5 ha (~80 trees, 40 banana mats, 40 yerba mate) + compound 440 m² (16 trees)	Surplus fruit feeds ducks and goats
Goats	7 does, 1 buck, 2 wethers + kids; 4 ha in 8 paddocks of 0.5 ha	Shed 48 m², buck pen 20 m²; ~1,280 L milk
★ Muscovy	26 hens + 6 drakes; ~110 table ducks a year	House 36 m²; pasture 1,600 m² in 4 cells; ~3,200 eggs expected (low end of the Muscovy range)
★ Fish	5 ponds × 500 m² = 2,500 m² + 250 m² duckweed	~187 kg fillet; one deep covered pond; 5 solar aerators
Cook weeks	4 a year, 91 days apart (Feb, May, Aug, Nov)	~403 portions, ~504 kg, ~630 L of flat frozen bags per cycle
★ Freezers	5 × 300 L = 1,500 L	~1,250 kWh/yr (~3.4 kWh/day)
★ Power	18 kWp (45 × 400 W), 30 kWh usable battery, 12 kW hybrid inverter, 10 kVA generator + ATS	Critical bus ~7.8 kWh/day, so ~3.8 days autonomy

Item	Figure	Notes
★ Water	Peak ~18 m³/day; rain tanks 80 m³; dam ~9,000 m³; well	Hard gates: well test and MADES approval
Cook-week labour	~500 kg of food in 6 days	Needs 2–3 helpers or a two-part cook week

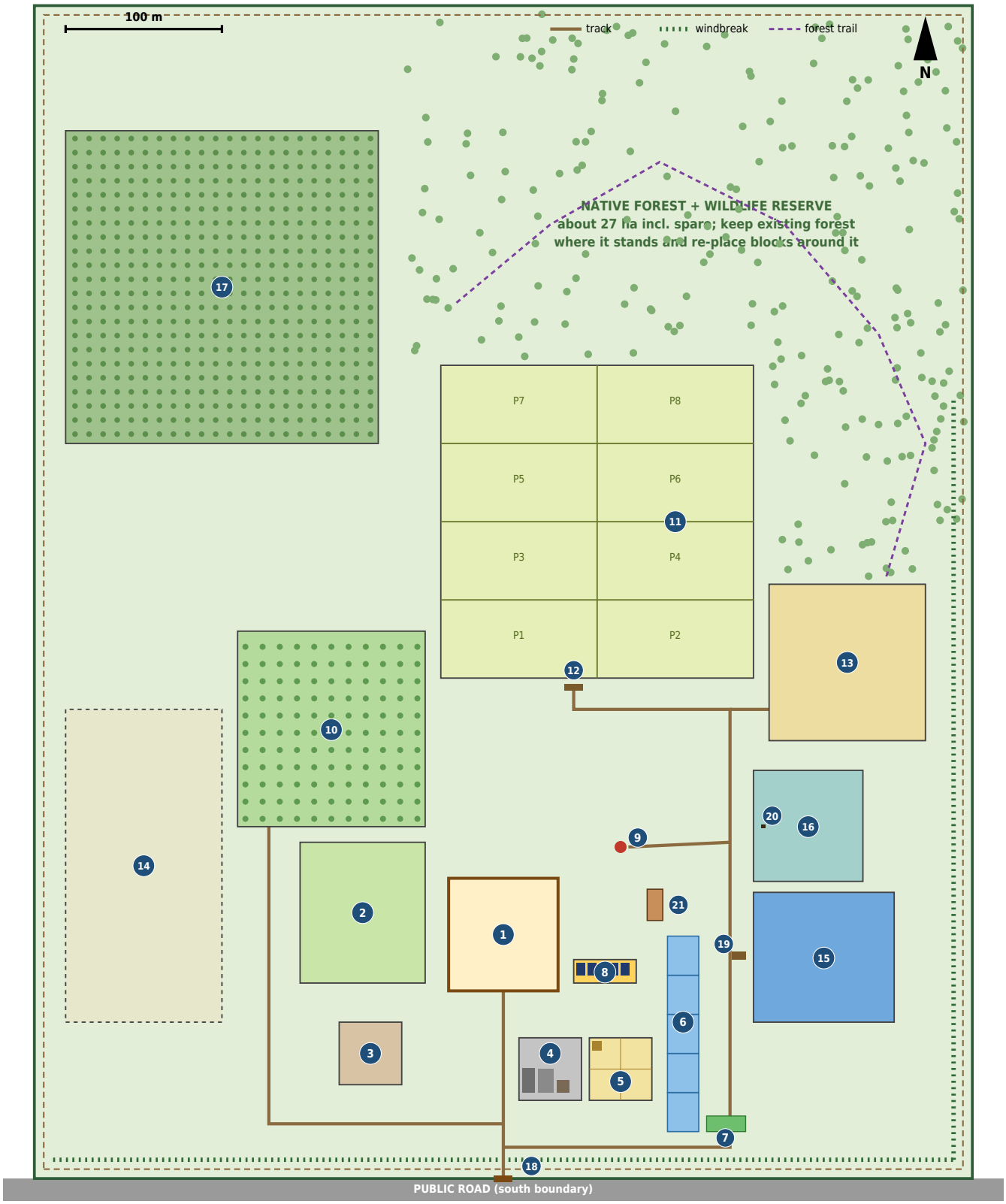
3. Land budget and farm layout (45 ha)

The plan keeps everything that needs daily attention within a short walk of the compound, and uses the extra land for things that need a visit every few days or less: grazing, orchards, grain, water storage, wood and wild habitat.

Block	ha	Size / notes
★ Walled compound + outer belts	0.6	Compound 5,040 m² plus outside planting and the entrance apron
★ Intensive food block	0.72	80 × 90 m: annual beds, shelters, perennials, shade house, mushroom room
★ Zone C (compost, humanure, vermicompost, firewood, biochar)	0.16	40 × 40 m, south-west of the compound (downwind of NE breeze)
★ Work barn complex (barn, hay/machinery shed, BSF shed, yard)	0.16	40 × 40 m on the entrance side
★ Muscovy duck system	0.16	40 × 40 m pasture, 4 cells, house 36 m²
★ Fish ponds (5) + duckweed	0.275	5 × 500 m² + 250 m²
★ Solar field, battery shed, generator	0.06	40 × 15 m beside the compound (18 kWp, ~90 m² of panels)
★ Main orchard	1.5	120 × 125 m
★ Goat silvopasture (8 paddocks of 0.5 ha)	4	200 × 200 m, 1,600 m of goat-proof fence
★ Grain fields (0.5 ha duck maize + 0.5 ha rotation)	1	100 × 100 m
★ Rotation and expansion reserve	2	Cover crop until needed; replaces the 1,020 m² reserve
★ Dam / reservoir	0.75	~9,000 m³; needs MADES approval and a hydrology check
★ Wildlife wetland	0.5	Shallow overflow wetland for wildlife; swimming is at the pool (or the dam, see section 6b)
★ Woodlot (firewood, charcoal, timber)	4	200 × 200 m
★ Tracks, firebreaks, windbreaks, fodder alleys	2	Perimeter track 2.7 km + internal tracks ~1.2 km + hedgerows
Allocated total	**17.89**	**39.7% of 45 ha**
Native forest, wildlife reserve, spare	**27.1**	Check the legal forest-reserve share with INFONA (a minimum share of forested land is commonly cited for properties over 20 ha; verify)

Not scaled up on purpose: goat herd, duck flock and the four cook weeks. Animals drive daily labour, and more land does not reduce it. The 2.7 km boundary gets a simple fence, gates and cameras at the entrance; the compound wall is the security line.

Farm site plan (to scale, 45 ha)



Concept only: assumes fall to the east, NE summer breeze, cold south wind, rectangular 600 x 750 m plot (45 ha). Survey before fixing.

Block schedule and legend for the site plan

#	Block	x, y (m)	Size (m)	Area m²	Notes
1	Walled compound	265, 120	70 × 72	5,040	South gate faces the entrance track; service gate east
2	Intensive food block	170, 125	80 × 90	7,200	West of the compound; staples, shelters, perennials, shade house, mushroom room
3	Zone C	195, 60	40 × 40	1,600	SW of compound; humanure, compost, vermicompost, firewood, biochar pad
4	Work barn complex	310, 50	40 × 40	1,600	Barn 125 m² (8 × 15.6), hay + machinery shed ~150 m², BSF shed 60 m²
5	Muscovy pasture + house	355, 50	40 × 40	1,600	Duck house at the NW corner, 10 m from the barn grain silos
6	Fish ponds 1–5	405, 30	20 × 125	2,500	5 × (20 × 25 = 500 m²); ponds 3–4 reached by gated duck path
7	Duckweed	430, 30	25 × 10	250	250 m²
8	Solar + battery + generator	345, 125	40 × 15	600	600 m² footprint
9	Well + pump house	372, 209	6 × 6	36	3 × 3 m house, final spot after the yield test
10	Main orchard	130, 225	120 × 125	15,000	~80 trees, 40 banana mats, 40 yerba mate
11	Goat silvopasture	260, 320	200 × 200	40,000	8 paddocks of 100 × 50 m
12	Goat hub (shed, milking, buck pen)	339, 312	12 × 4	48	South edge of the goat block
13	Grain fields	470, 280	100 × 100	10,000	Duck maize + rotation
14	Rotation + expansion reserve	20, 100	100 × 200	20,000	Cover crop until needed
15	Dam	460, 100	90 × 83	7,470	~7,470 m² surface, ~9,000 m³
16	Wildlife wetland	460, 190	70 × 71	4,970	~4,970 m²
17	Woodlot	20, 470	200 × 200	40,000	4 ha
18	Farm entrance gate	294, -2	12 × 4	48	5 m gate, camera, intercom, post box, 60 m² parking
19	Dam-side shelter + fishing deck	446, 140	8.4 × 5	42	30 m² picnic shelter + 12 m² deck, fire ring on a gravel apron
20	Wetland bird hide	465, 224	2.5 × 2	5	5 m² hide on the wetland edge
21	Multi-sport court	392, 165	10 × 20	200	200 m² fenced court (5-a-side, basketball, tennis-style), solar LED lights

Not numbered on the map: the ~3 km forest trail (dashed purple) with 2 benches, and the optional 8 × 8 m camp pad in the woodlot clearing.

x, y are metres from the south-west corner (x east, y north). Block areas are the drawn rectangles; ponds are 5 rectangles of 500 m². Everything not listed is windbreak, track, fodder alley or native forest.

Walking distances (straight line, from the east mudroom door at about 313, 144 m)

Chore point	From mudroom door (m)	Walk at 1.3 m/s (min)	Use a vehicle?
Compound main gate	27	0.3	No
Solar + battery	53	0.7	No
Mushroom room (food block east edge)	63	0.8	No
Work barn	76	1.0	No
Duck house	75	1.0	No
BSF shed	90	1.2	No
Well + pump house	92	1.2	No
Food block centre	106	1.4	Optional
Zone C compost	117	1.5	Optional
Ponds (centre)	113	1.5	Optional
Goat hub (milking)	173	2.2	Yes
Main orchard centre	189	2.4	Yes
Farm gate	145	1.9	Optional
Dam centre	192	2.5	Yes
Grain block centre	278	3.6	Yes

Goats are the one daily chore beyond a 2-minute walk. That is the price of putting the well about 100 m from the goat shed and keeping goat smell away from the veranda. Use the utility vehicle for feed days, or move the goat hub closer if the survey allows the well to move.

Why the blocks sit where they do

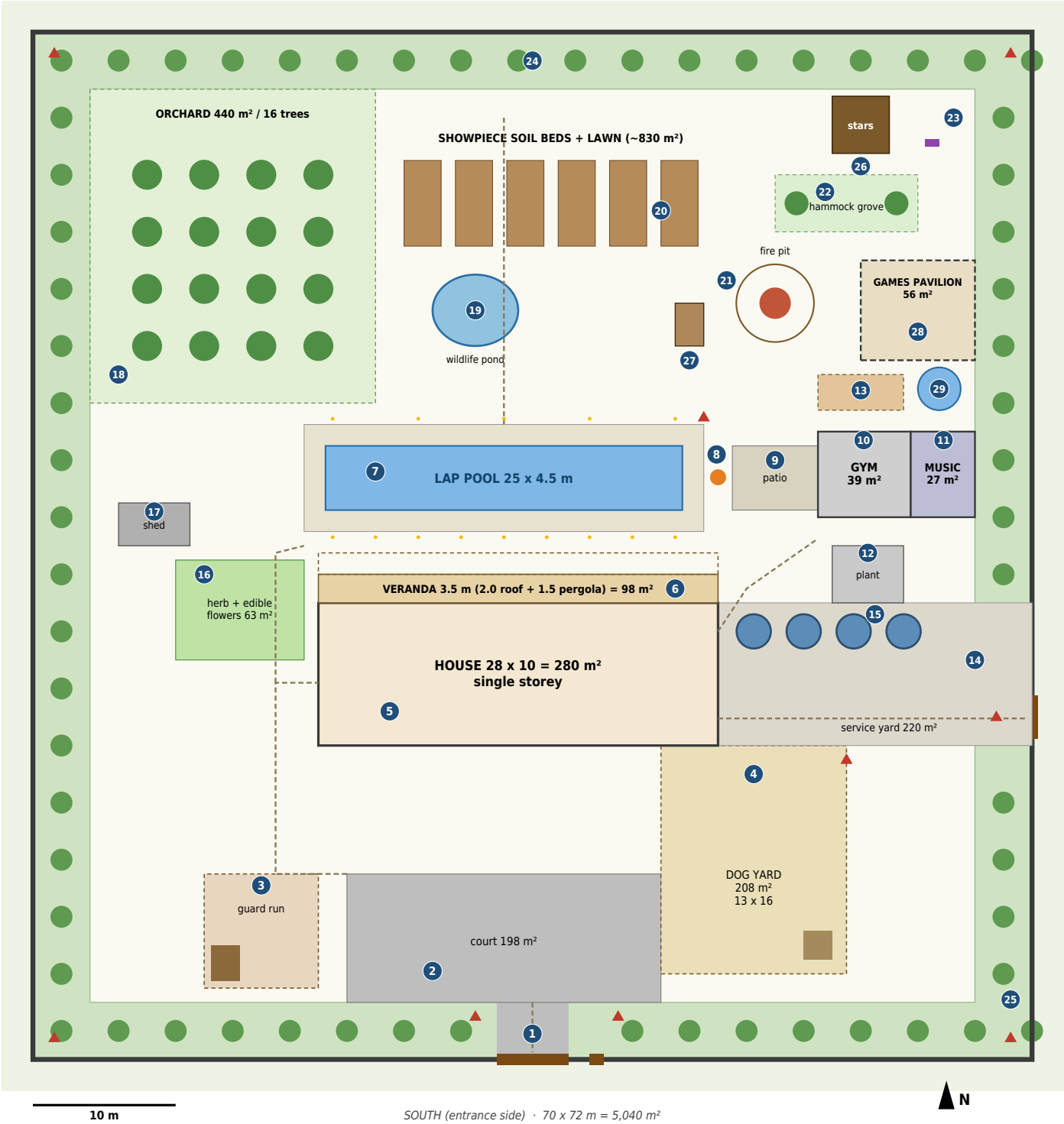
- **Smell and wind:** compost, humanure and firewood (Zone C) are south-west of the compound, so the assumed NE summer breeze carries smell away. The BSF shed sits at the barn, where slaughter waste is produced.
- **Clean and dirty:** clean side is the north (garden, veranda, pool). The barn, ducks, ponds and Zone C are south and east. Trucks, slaughter and fuel never pass the veranda: they use the east service gate.
- **Water and slope:** ponds, dam and wetland sit on the assumed low east side. Bed runoff and pool drain water never reach ponds or the well. A solar pump lifts water to a header tank on the high west side for gravity drip.
- **Well:** about 100 m from the goat shed, 120 m from the duck house, 150 m from the barn and about 200 m from Zone C. It is about 75 m from the nearest pond (rule: at least 50 m from ponds).

- **Goat wind caveat:** the goat block sits north-north-west of the compound. If the survey shows wind mostly from the north or north-east, swap the goat block with the grain fields.

4. Compound design (70 × 72 m, 5,040 m², walled)

Design idea: from the road you see a wall and trees. Inside, a quiet south-to-north sequence runs from gate and dogs, through the house and veranda, across the pool, to a garden and orchard. Everything noisy or messy is at the edges or outside the wall.

Plan to scale (north up)



Compound plan legend

#	Element	Size / notes
1	Main gate 5 m motorised sliding + 1 m pedestrian gate with intercom	South wall, centre. Drive runs 4 m through the belt to the court.
2	Parking and turning court	22 × 9 = 198 m², permeable gravel or pavers
3	Guard-dog kennel + run	8 × 8 = 64 m² run + insulated 2 × 2.5 m kennel, west of the court so the dogs cover the gate
4	Companion dog yard	13 × 16 = 208 m², shaded rest platform, dig pit 4 m², tap; dog door from the mudroom; self-closing gate
5	House	28 × 10 m = 280 m², single storey
6	North veranda	28 × 3.5 = 98 m²: inner 2.0 m solid roof, outer 1.5 m deciduous pergola
7	Lap pool + decks	Pool 25 × 4.5 m; pool and decks 28 × 7.5 = 210 m²; cover, fence, alarm, dog ramp
8	Outdoor solar rinse shower + towel bench	Between pool end and patio
9	Covered patio	6 × 4.5 = 27 m² between pool and gym doors
10	Gym	6.5 × 6.0 = 39 m²
11	Music room	4.5 × 6.0 = 27 m², sealed
12	Pool plant room	5 × 4 = 20 m²: pump, filter, salt cell, controls on the dashboard
13	Sauna + cold plunge (optional)	6 × 2.5 = 15 m²
14	Service gate 3 m + service yard	22 × 10 = 220 m²; gate in the east wall at the mudroom door; harvest, barn runs, fuel, tradespeople
15	Rain-tank bank	4 × 10 m³, screened by planting
16	Herb + edible flower garden	9 × 7 = 63 m², beside the potting room
17	Garden shed	5 × 3 = 15 m²
18	Compound orchard	20 × 22 = 440 m², 16 trees, greywater mulch basins
19	Wildlife pond + bat and bird boxes	~30 m², stocked with small native fish against mosquitoes
20	Showpiece soil beds + lawn	~830 m²; compost vs mulch vs biochar test beds, soil journal, sensors
21	Fire pit + seating	~50 m² at the garden's east edge, view over beds and pool
22	Hammock / daybed grove	10 × 4 = 40 m² under trees, good airflow
23	Quiet nook / thinking bench	NE corner, view back over the garden
24	Tree belt	4 m wide inside all four walls = 1,072 m²: layered native and fruit/flower planting
25	Cameras (9 on the plan; 8 minimum)	Four corners, both gates, dog yard, pool and service yard approach
26	Stargazing deck	4 × 4 = 16 m² raised timber deck, telescope pad, red-light only
27	Outdoor cooking and pizza table	2 × 3 = 6 m² with grill grate for the fire pit
28	Games pavilion	8 × 7 = 56 m², roofed and open-sided: pool table, table tennis, darts, bar counter, fans, drop-down screens
29	Hot tub	3 × 3 = 9 m² footprint, solar or wood heated, beside the sauna

Area ledger (5,040 m²)

Element	m²	Element	m²
Perimeter belt (4 m)	1,072	Herb garden	63
Court	198	Fire pit	50
Guard run + kennel	80	Hammock grove	40
Dog yard	208	Sauna + plunge	15
House	280	Shed	15
Veranda	98	Wildlife pond	30
Annex + patio	93	Tank bank	30
Service yard	220	Stargazing deck	16
Pool + decks	210	Cooking table	6
Plant room	20	Games pavilion	56
Showpiece beds + lawn	830	Hot tub	9
Orchard	440		
Named total	**4,079**	**Paths, drive, planted borders, spare**	**961**

Named elements come to 80.9% of the compound. The remainder is paths, driveway, borders and growing room. If budget forces cuts, trim the sauna, the cold plunge and the outer belt width before touching the pool, gym or orchard.

How the compound works

Topic	Design
Daily flow	Gate, court, house. Morning: gym, patio, pool, rinse shower, house (no wet walk through the house). Evening: veranda, fire pit, stars, TV or guitar.

Topic	Design
Harvest and trucks	Cook-week harvest, fuel and deliveries use the east service gate and a straight path to the mudroom door. One-way flow: barn, service gate, mudroom, processing, freezers, pantry, kitchen.
★ Security	2.4 m solid wall (284 m), motorised gate, intercom, 8–9 cameras with phone alerts, motion lighting on the wall line, dogs at the gate, tree belt as a visual screen, alarm panel with siren and panic buttons, fire extinguishers at every door. The farm boundary is a simple fence; the wall is the real line.
★ Lighting	Warm, low, shielded lighting only: veranda, paths, pool edge, fire pit. Motion lights at gates and wall corners. No glare toward neighbours or wildlife.
★ Pets	Companion dogs live in the house and the 208 m² yard; guard dogs at the gate run. Dog wash and dog door at the mudroom, pool ramp, self-closing gates, shaded rest platforms. Dogs are kept apart from ducks and goats until trained; a livestock-guardian dog is bonded to the goats from puppyhood. Optional 6 m² catio at the veranda's west end if you keep cats.
★ Nature	Wildlife pond, bat and bird boxes, native flowering shrubs and climbers for scent and insects, 16-tree orchard, herb garden, hedgerow-style belts. Choose native, low-maintenance species and avoid invasive ornamentals.
Greywater	Showers, laundry and sinks (~300 L/day) to mulch basins in the compound orchard via a grease trap. Never to the pool, ponds or well.
Private by design	Built first for your own daily enjoyment, not for entertaining, and hosting is welcome whenever it benefits you and your partner personally: the veranda, fire pit, pool, games pavilion, patio and bedroom 3 all work for friends or family. No formal front garden; one main gate, one service gate, no view in from outside.

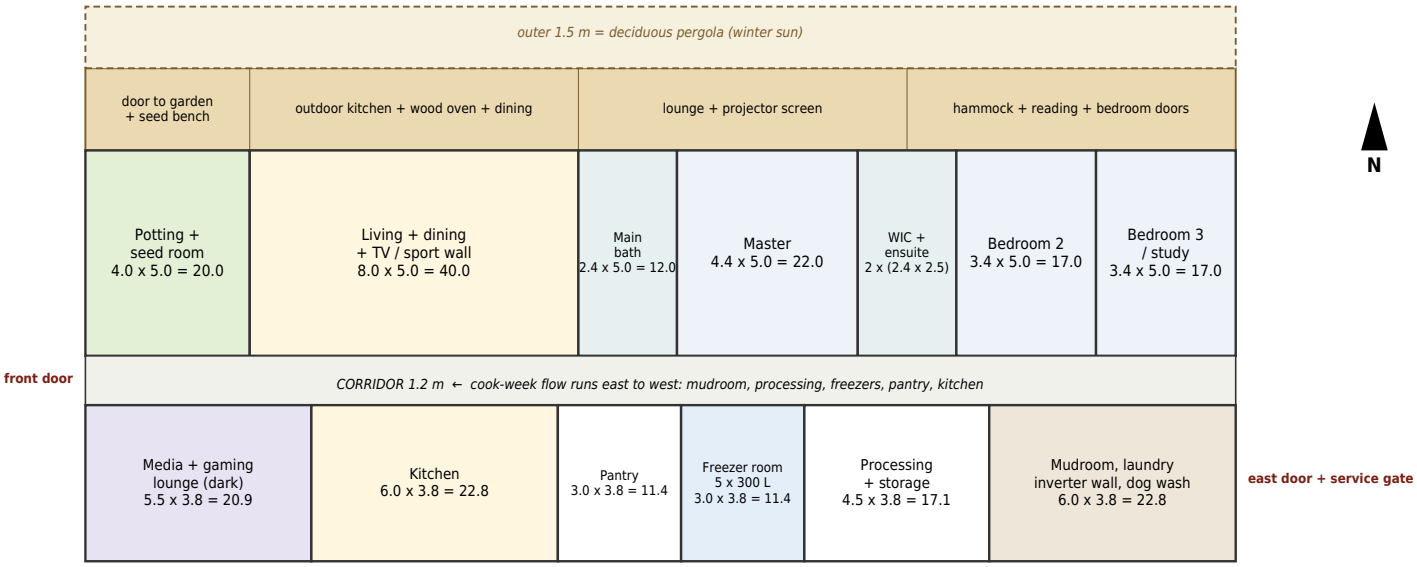
Compound orchard (16 trees)

4 citrus (lemon, lime, mandarin, orange), 3 guava, 2 avocado (cold-hardy), 2 dwarf mango, 2 pitanga, 1 jabuticaba, 1 acerola, 1 fig, planted on a ~4 m grid. Banana mats sit in the greywater basins along the west belt. Pollinator strips run between the rows.

5. House design (28 × 10 m, 280 m², single storey)

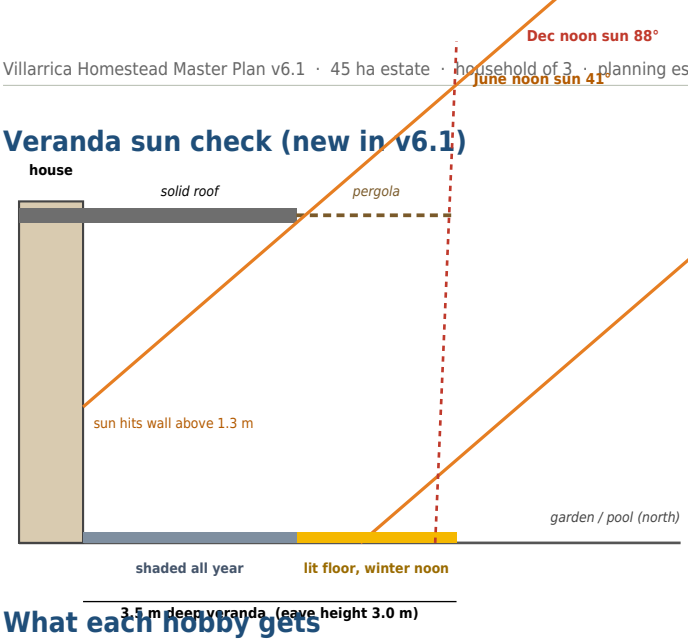
Rooms are ordered by noise and mess, with the garden and veranda on the north (sun, view, clean side) and the work rooms on the south-east, next to the service gate. The bath sits between the living room and the master bedroom as a sound buffer; the media lounge and potting room are at the far west end, away from sleeping.

Floor plan to scale



Room	Size (m)	Area m²	Role and placement
★ Potting + seed room	4.0 × 5.0	20.0	West end; door to the garden and herb beds; sink, bench, seed starting; the one room allowed to be muddy
★ Living + dining + TV / sport wall	8.0 × 5.0	40.0	Opens to the veranda; casual TV and sport wall (75 in), sofas facing the garden, dining for 4–6, wide opening to the kitchen
★ Main bath	2.4 × 5.0	12.0	Sound buffer between living and master; shared by bedrooms 2 and 3
★ Master	4.4 × 5.0	22.0	Quiet middle, garden view, door to the veranda
★ WIC + ensuite	2 × (2.4 × 2.5)	12.0	Walk-in wardrobe 2.4 × 2.5 and ensuite 2.4 × 2.5, doors from the master only
★ Bedroom 2	3.4 × 5.0	17.0	Bedroom (door to veranda)
★ Bedroom 3 / study	3.4 × 5.0	17.0	Bedroom or study; second desk for remote work
★ Media + gaming lounge (dark)	5.5 × 3.8	20.9	South side (no glare), blackout blinds, 120 in projector or 85 in TV, 5.1 sound, acoustic panels, gaming PC and console desk, recliners; the movie, series, sport and gaming room
★ Kitchen	6.0 × 3.8	22.8	Cooking is a hobby: induction hob + oven, two sinks, long work bench, pass-through to the living room, pantry next door, door to the veranda cooking bay
★ Pantry	3.0 × 3.8	11.4	Between kitchen and freezers; dry goods for the week, cleaning closet
★ Freezer room 5 x 300 L	3.0 × 3.8	11.4	5 chest freezers (1,500 L), own ventilation, temperature probes on the alarm network, beside the inverter wall
★ Processing + storage	4.5 × 3.8	17.1	Vacuum sealing, portioning, bagging; harvest enters from the mudroom
★ Mudroom, laundry inverter wall, dog wash	6.0 × 3.8	22.8	East entrance by the service gate; boots, laundry, dirty shower, dog wash, dog door to the yard, inverter wall, controller node
Corridor	28.0 × 1.2	33.6	Every room opens onto it; cook-week flow runs east to west
Total house	**28.0 × 10.0**	**280.0**	Rooms 246.4 + corridor 33.6
Veranda	28.0 × 3.5	98.0	Primary living room; inner 2.0 m roofed (56 m²), outer 1.5 m pergola (42 m²)

Veranda sun check (new in v6.1)



Result: at about 25.8°S the sun at noon is ~41° above the horizon on 21 June and ~88° on 21 December. A solid 3.5 m roof at 3.0 m height would shade the entire floor in winter (the shadow reaches 3.0 / tan 41° = 3.5 m), which is the opposite of what v5.2 assumed. Fix: a 2.0 m solid inner roof (shade, dining, rain cover) plus a 1.5 m outer pergola with deciduous vines or adjustable louvres, so the outer strip is sunny in June and shaded in summer. Winter noon sun also reaches the wall above ~1.3 m, warming the living room glass. Check again with the real orientation and roof height after the survey.

What each hobby gets

You like	Where it lives	What it has
★ Chilling and resting	Veranda 98 m²; hammock grove 40 m²; fire pit; master suite	Hammock and daybed bay, sofa bay, outdoor dining bay; ceiling fans; shade from the pergola in summer; fire pit with seating for evenings; nook bench in the NE corner.
★ TV, series, sport, gaming	Living-room TV wall 40 m²; media lounge 20.9 m²; veranda projector bay	Casual viewing in the living room; dark, treated media lounge for films, sport and gaming (projector or 85 in screen, 5.1 sound, wired Ethernet); outdoor screen on summer nights; Starlink plus mobile backup; PC and console desk. The media lounge is the first thing to cut if budget is short.
★ Cooking	Kitchen 22.8 m²; pantry 11.4 m²; veranda cooking bay; processing 17.1 m²	Induction and oven, two sinks, long bench; wood-fired oven (~2 m²) and braai on the veranda for personal use; herb garden and potting room within 10 m; processing room and five freezers for cook weeks.
★ Guitar	Music room 27 m² in the annex; veranda corner for unplugged playing	Double-leaf sealed door, double wall with air gap, optional floating floor, acoustic panels, amp bay, own split AC; 13 m from the nearest bedroom.
★ Gym	Gym 39 m² (6.5 × 6.0 m), ceiling 3.2 m	Power rack and bench rated for heavy loads, barbell and ~300 kg of plates, dumbbells to 50 kg, cable/lat station, rower or bike, 20 mm rubber floor over a 150 mm reinforced slab, mirror wall, two big doors to the patio, two ceiling fans.
★ Swimming	Lap pool 25 × 4.5 m; patio; rinse shower	Gym, patio, pool, rinse shower, house in one line; cover, fence, alarm, dog ramp; swim season about October to April.
★ Nature	Garden, orchard, wildlife pond, belts	Potting room opening to the beds, herb and edible-flower garden, showpiece soil beds, 16-tree orchard, wildlife pond, bat and bird boxes, screened windows on both faces of every room.
★ Pets	Dog yard 208 m²; guard run 64 m²; mudroom	Dog door from the mudroom, dog wash, shaded platforms, dig pit, tap, pool ramp, self-closing gates, separate guard run at the gate; livestock-guardian dog later.

Building notes

- Single storey, polished or sealed concrete floor, block or brick walls, insulated metal roof with deep eaves and ridge vents; 3.2 m ceilings; cross-ventilation (windows on opposite faces of every room); ceiling fans in all rooms; split AC only in the master, media lounge and music room.
- Screens on every opening (dengue and insects); double glazing on the media lounge and music room.
- Solar hot water (2 × 200 L) with electric backup; UV plus carbon filter on the drinking-water tap; whole-house surge and lightning protection; Cat6 cable to every room.
- Wet areas stack where possible (bath, ensuite, kitchen, laundry) so pipe runs are short and repairable. Greywater leaves by a separate line to the orchard.
- Rain-tank bank (4 × 10 m³) at the east end in the service yard, with first-flush diverters.

Phasing and cut list

Phase	Build	Area	Notes
Core (first)	Living, bath, master suite, bedroom 2, kitchen, pantry, freezer room, processing, mudroom, corridor	222.1 m² + veranda	Runs the whole cook-week system
West wing	Potting room + media lounge	40.9 m²	A desk in the living room covers the gap
Bedroom 3	Third bedroom or study	17.0 m²	Build when the third adult arrives; the pergola can follow
Annex	Gym + music room + patio	66 + 27 m²	Resident priority: build early if budget allows
Easiest cuts	Media lounge (use the living-room wall), then patio roof, sauna and cold plunge		Gym, music and pool are not cuts

6. Annex, pool and recovery zone

Item	Plan
★ Annex size and position	11.0 × 6.0 m = 66 m² plus 27 m² covered patio on its west side (6.0 × 4.5), north-east of the house, joined by a covered walkway from the east end of the veranda. Power and network from the house in the same trench. On the non-critical bus (shed first).
★ Gym (39 m²)	Sized for a heavy lifter: rack, bench, plates, dumbbells, cable station, cardio, mirror wall, 20 mm rubber floor on a reinforced slab. Doors open to the patio for airflow; fans plus a dehumidifier option for the rainy season. Opens straight onto the patio and pool end.
★ Music room (27 m²)	East half of the annex, farthest from the house. Double-leaf sealed door, double wall with an air gap, acoustic panels, own split AC, amp bay. Guitar stands on the wall; a veranda corner is the unplugged spot.
Lap pool	25.0 × 4.5 m (112.5 m²), long axis east-west in front of the veranda, depth 1.2–1.5 m (~150 m³); 1.5 m deck each side, 28 × 7.5 m footprint. Visible from the veranda, living room and bedrooms.
Daily flow	Gym, patio, pool, outdoor rinse shower, house: train, swim, shower, recover with no wet walk through the house.
Water	First fill ~150 m³ from the dam or well in the wet season. Summer evaporation estimated ~0.6 m³/day (112.5 m² × ~5 mm); cover when not in use.
Power	Variable-speed pump ~0.5 kW × ~6 h = ~3 kWh/day (estimate), solar-direct in daytime, on the non-critical bus, first load shed in any power shortfall.
Treatment	Cartridge or sand filter with salt chlorination (proposal), 5 × 4 m plant room with pump, filter and controls on the dashboard.
Season and heating	No heating in the baseline; solar cover extends the season. Expect about October to April (assumption). A solar mat (~40 m² of collector) is an optional upgrade for winter swims.
Drain and backwash	To a soak pit near the service yard, at least 50 m from the well and ponds, never to beds. Pool water and chlorine stay out of the food system.
Safety	Fence with self-closing gate on the dog side, pool alarm, cover, non-slip deck, dog ramp; a phone check-in alert on the dashboard when swimming alone; first-aid kit and life ring at the patio.
★ Sauna + cold plunge (optional)	6 × 2.5 m = 15 m² beside the annex; wood-fired or electric (electric on the non-critical bus). Cold plunge uses pool or tank water, drained to the soak pit.
Cost	TBD (quote locally; a shorter pool or swim spa is the fallback if budget or water is short).

Swimming elsewhere: the fish ponds and the wetland stay out because of duck manure and fish; swimming in the dam is a possible option once the water tests clean, with a gradual-entry ramp and a safe edge at the shelter (see section 6b). The pool is the everyday place.

6b. Entertainment and leisure layer (new in v6.1)

Everything here is chosen for your own daily enjoyment, not for entertaining, and it works well whenever you choose to host friends or family. Items are picked because they get used weekly, cost little power and little upkeep, and sit on the non-critical bus (shed first). Sizes are planning figures.

Item	Where	Size / quantity	Notes
★ Outdoor sound zones	Veranda, pool deck, patio, fire pit	6 weatherproof speakers, one amplifier in the media lounge, phone-controlled zones	~0.2 kW when playing; turn it down at night if you like (animals and wildlife sleep nearby)
★ Garden cinema	Veranda east gable, lawn seating	3 m pull-down screen, shared projector with the veranda bay, 4 loungers	~0.3 kW; summer nights and big matches
★ Pool-side comfort	Pool deck	4 sun loungers, 20 m² shade sail, warm/white pool and edge lighting	Lighting ~0.1 kW
★ Games and reading nook	Veranda west bay by the potting room	Table for cards and board games, 3 m bookshelf, daybed	Weather-protected by the 2.0 m roofed band
★ Coffee and tea corner	Veranda cooking bay	Small drinks fridge + kettle and coffee station	~0.1 kW
★ Offline media server (NAS)	Media lounge	20 TB with films, series, music and games	Keeps entertainment working when Starlink drops; ~30 W
★ Games pavilion + bar	East garden (item 28)	8 × 7 = 56 m², roofed, open-sided: pool table, table tennis, darts, bar counter, drinks fridge, fans	Fully usable in rain; ~0.4 kW lights and fans
★ Hot tub	Beside the sauna (item 29)	3 × 3 m footprint, seats 4	Solar or wood heated, electric only as backup; drained to the soak pit
★ Multi-sport court	East of the compound, near the solar field (21)	10 × 20 m = 200 m², fenced, 5-a-side, basketball, tennis-style	Solar LED lights; about 100 m from the mudroom
★ Recording corner	Music room	Small desk, interface and monitors	Record guitar ideas without leaving the property
★ Stargazing deck	NE corner of the compound (item 26)	4 × 4 m = 16 m², telescope pad (0.6 × 0.6 m pier), red-light only	All compound lights are warm, low and shielded, so the sky stays dark
★ Outdoor cooking table	Fire pit (item 27)	2 × 3 m = 6 m² prep table + grill grate	The wood-fired oven stays on the veranda
★ Run and cycle loop	Perimeter track	2.7 km × 4 m, shaded in places, 2 water points	3 bicycles or e-bikes; dogs can run it with you
★ Forest trail	Native reserve (dashed on the map)	~3 km × 1.2 m, 2 benches, left unmanaged otherwise	Four wildlife cameras feed the dashboard
★ Wetland bird hide (20)	Wildlife wetland west edge	2.5 × 2 m = 5 m²	Binoculars and a field guide; no boats on the wetland

Item	Where	Size / quantity	Notes
★ Dam-side shelter + fishing deck (19)	Dam west shore	30 m² picnic shelter + 12 m² deck, fire ring on a gravel apron	Fish species per MADES; optional swimming steps after a water-quality test; fire ring 6 m clear of vegetation
★ Kayaks (optional)	Dam only	2 kayaks	Not on the wetland or ponds
★ Camp pad (optional)	Woodlot clearing	8 × 8 m, fire ring, 2 hammock posts	Sleeping out; inside the firebreak rules
★ Hammocks	Grove, veranda, camp pad	4 hammocks	Cheap and used daily
★ Dog play	East lawn of the compound, loop track	~100 m² of lawn kept free, pool ramp, shaded water points	Optional agility corner
★ Wildlife watching	Compound and belts	6 bird feeders, nest and bat boxes, pollinator strips	Species chosen native; feeders cleaned weekly

Priority	What	When
1: with the house	Cable and conduit for sound and projector, loungers, NAS, bikes, hammocks, bird feeders, pizza and cooking table	House and compound build (6–18 months)
2: cheap and soon	Games pavilion, hot tub, stargazing deck, loop-track water points, forest trail, bird hide, pool-side shade sail, games nook	12–30 months
3: after year 2	Multi-sport court, dam shelter and fishing deck, camp pad, kayaks, sauna and cold plunge	24–36 months, once the dam holds water

Defaults, not restrictions: the pool stays the everyday swimming place; dam swimming is optional after a water test; warm shielded lighting keeps the stargazing deck dark; fire rings sit on gravel with firebreaks for safety; quiet hours are your call. Hosting is welcome whenever it is good for you and your partner. Cost is TBD.

7. Food: needs, harvest calculation and nutrients (3 adults)

Method: the v5.2 food-need list was scaled by ×1.12 so that its energy matches the 3-person target, then the +30% margin was applied (eggs excepted). Harvest = area × yield per m² per year (yields taken from the v5.2 supply table, all untested). Annual crops are planted at 1.5× the minimum area because double sowing, chia and soil-bed yields are unproven.

Person	Energy kcal/day	Protein g/day	Share of energy
Man (target 125 kg, building muscle)	4,400	245	50%
Woman 1	2,200	100	25%
Woman 2	2,200	100	25%
Household	**8,800**	**445**	100%

Food	v5.2 list (per year)	3-person need ×1.12	kcal/day	Protein g/day
Cassava	179	200	879	8
Sweet potato	241	270	636	12
Maize	102	114	1,142	29
Cowpea	99	111	1,021	71
Peanuts	22	25	383	17
Sunflower seeds	22	25	394	14
Pumpkin	113	127	90	3
Chia	15	17	224	8
Greens, tomato, pepper	329	368	222	15
Eggs (each)	2,300	2,576	494	44
Milk (L)	730	818	1,546	81
Goat cheese	11	12	101	7
Fish fillet	117	131	345	72
Meat (duck, goat)	128	143	550	86
Mushrooms	55	62	37	5
Honey	15	17	140	0
Fruit (edible)	186	208	342	5
Avocado flesh	57	64	280	3
Total from the need list			**8,826**	**482**
Target			**8,800**	**445**
Coverage of target			**100%**	**108%**

Units: kg per year, except eggs (count) and milk (litres). Composition values are ballpark per kg of edible product (for example maize 3,650 kcal and 94 g protein; goat meat and duck blended at 1,400 kcal and 220 g; goat milk 690 kcal and 36 g per litre) and carry about ±20% error. The 329 kg "greens, tomato, pepper" line includes ~24 kg not itemised in v5.2. The v5.2 list alone gives ~7,880 kcal and ~430 g (about 90% and 97% of target); I earlier quoted ~420 g, which was a rougher sum.

Plant harvest calculation

Crop	Harvests / yr	kg/m²/yr	Min m²	Planted m²	Harvest kg (min / planted)	Need kg	Margin (min / planted)	kg per harvest (planted)
Cassava	1 (4 rolling blocks)	0.97	270	405	262 / 393	200	+31% / +96%	98.2
Sweet potato	3 batches	1.28	275	410	352 / 525	270	+30% / +94%	174.9
Maize (people)	2 sowings	0.6	250	375	150 / 225	114	+31% / +97%	112.5
Cowpea	2 sowings	0.37	390	585	144 / 216	111	+30% / +95%	108.2
Peanuts	1	0.25	130	195	32 / 49	25	+32% / +98%	48.8
Sunflower seed	2 sowings	0.078	410	615	32 / 48	25	+30% / +95%	24.0
Pumpkin	2	2.42	70	105	169 / 254	127	+34% / +101%	127.0
Chia + perilla	1 (Apr–May)	0.08	275	410	22 / 33	17	+31% / +95%	32.8
Cooked greens	4 bulk cuts	1.79	80	120	143 / 215	106	+35% / +102%	53.7
Salad + herbs	weekly basket	3.12	28	40	87 / 125	67	+30% / +86%	2.4
Tomato + pepper	weekly picks	3.25	70	105	228 / 341	168	+35% / +103%	13.1
Total area			**2,248**	**3,365**				

"kg per harvest" divides the planted-area yield by the number of harvests (for cassava, one of four blocks is dug per cook week; for sweet potato, one of three batches). The 1.5× planting is deliberate insurance, and surplus goes to goats (vines, leaves, pumpkin) and ducks. If double sowing and chia pass their trials, the planted area can shrink toward the minimum.

★ Item	Size / quantity	Notes
Rain shelters (greens, salad, tomato, pepper)	6 × 45 m² = 270 m²	Open-sided, vents with temperature and humidity sensors; covers the 265 m² of planted beds for these three crops

★ Item	Size / quantity	Notes
Perennial staples	~800 m²	Taro, perennial spinach, pigeon pea, moringa, sweet-potato vine reserve; not counted toward the targets (bonus and goat fodder)
Shade house + seedling tunnel	~100 m²	Seedlings, cuttings, frost cloth, temperature-controlled vents
Mushroom room	25 m² (5 × 5)	Bags on shelves, climate sensors; ~87 kg expected (v5.2 ratio 70 kg per 20 m²)
Vermicompost bay + biochar kiln pad	~6 m² + ~30 m²	In the food block next to Zone C route
Paths, headlands, hedges, spare	~1,800 m²	Food block is 7,200 m² in total

Animal products, orchard and other foods

Food	3-person need	Target (+30%)	Plan	Expected	Margin vs need
Eggs	~2,600	margin relaxed	26 Muscovy hens + 6 drakes	~3,200 (low end of 100–150 per hen)	+24%
Milk + cheese	818 L drink + 12 kg cheese (~123 L) = ~941 L	~1,223 L	7 does in milk, at least 4 in milk all year	~1,280 L (183 L per doe)	+36%
Fish fillet	131 kg	170 kg	5 ponds × 500 m², tilapia + traira	~187 kg (150 kg per 4 ponds × 1.25)	+43%
Meat	143 kg	186 kg	~110 table ducks + ~8 wether kids	~193 kg	+35%
Mushrooms	62 kg	80 kg	25 m² room	~87 kg	+41%
Fruit (edible)	208 kg	270 kg	Compound orchard + main orchard	Mature orchards give far more; the surplus feeds animals	large
Avocado flesh	64 kg	83 kg	2 compound + 12 main orchard trees	Yields from year 5–7	large (late)
Honey	17 kg	22 kg	8–10 hives + 2 stingless boxes	~40–60 kg possible (v5.2: 20 kg from 4 hives)	large

Main orchard (1.5 ha): about 80 trees (15 citrus, 12 guava, 12 avocado, 8 mango, 15 papaya, 10 pitanga, 6 mulberry), 40 banana mats and 40 optional yerba mate under the trees. Banana and frozen guava cover the September–November fruit gap. Plant in the first wet season: trees take 3–7 years to full output.

Nutrient gaps and checks for three people

- **Energy and protein:** need-list foods cover 100% of energy and ~108% of protein; the cooked-plus-fresh plan in section 8 delivers about +8% energy and +17% protein over target, which is the working buffer. The man's 245 g protein is covered by meat, fish, eggs, milk and cowpea.
- **Vitamin D:** midday sun, sun-dried mushroom gills, eggs, fish. Test 25-OH vitamin D for all three in winter and supplement if low.
- **EPA + DHA:** about half of target comes from food; algae oil ~250 mg/day for each woman, and for the man if the omega-3 index is low.
- **Iodine:** iodized salt is the only reliable source (~150 µg per adult per day); check the label's iodine content. Non-negotiable.
- **ALA:** most comes from chia (17 kg/yr need). Chia is untested; perilla is the backup; if both fail ALA falls below target, so the chia trial matters more with three people.
- **Iron, vitamin A, selenium (the man):** can run high; ferritin test once, no liver, avoid extra supplements with these. Women's iron stays well above target.
- **Calcium, B12, choline:** from milk, cheese, eggs, meat and fish; verify in the full re-run.
- **Folate:** folic acid supplement if pregnancy is planned. **Fibre** is high (beans, seeds, roots): build up gradually.
- **Scope:** macros are scaled and checked line by line; micronutrients are scaled but not re-run item by item. Do the full nutrient re-run after the first full year of egg, milk and yield data.

8. Cook weeks, meals, stored versus fresh food

Four cook weeks a year (Feb, May, Aug, Nov), 91 days apart. Rule: freeze as many cooked meals as possible; fresh harvest only for the weekly basket. A “portion” is a 1.25 kg bag (about 1.1 portions per slot per day across three people).

Slot	Meal	v5.2 per 91 d	v6.1 per 91 d	kg	Reheat
★ Breakfast	Sopa paraguaya (maize, cheese, onion, egg, milk)	46	51	64	Oven 15 min
★ Breakfast	Mushroom-egg tortilla squares (+ moringa)	45	50	62	Pan 5 min
★ Lunch	Goat-bean-greens stew with cassava, pumpkin	39	43	54	Simmer 15
★ Lunch	Duck locro (maize, beans, pumpkin)	26	29	36	Simmer 15
★ Lunch	Duck or goat curry with sweet potato, moringa, chilli, tomato	26	29	36	Simmer 15
★ Dinner	Baked tilapia, tomato-pepper sauce, sweet-potato mash	52	57	71	Oven 20
★ Dinner	Tilapia + sweet-potato fish cakes	39	43	54	Pan 8
★ Snack	Chipa (cassava starch, cheese, egg)	46	51	64	Oven 10
★ Snack	Peanut, sunflower, chia, honey, banana bars	45	50	62	Thaw 10
Total		**364**	**403**	**504**	~630 L of flat bags

Flavour bases (tomato-pepper sauce, curry paste, moringa-peanut pesto) are made at cook week with bone broth. A duck and pumpkin hash can replace a locro or curry once its portions are set. Eggs: ~1,700 a year go into meals (~425 per cycle); ~900 stay fresh. v5.2 counted both 575 per cycle in meals and 3–4 fresh eggs a day, which double counts; the egg-light recipes (starch or mash as binder) make the split work.

Seven-day eating rotation

Day	Breakfast	Lunch	Dinner	Snack
Mon	Sopa paraguaya	Goat-bean-greens stew	Baked tilapia + sweet-potato mash	Chipa
Tue	Mushroom-egg tortilla	Duck locro	Tilapia fish cakes	Seed bars
Wed	Sopa paraguaya	Duck or goat curry	Baked tilapia + sweet-potato mash	Chipa
Thu	Mushroom-egg tortilla	Goat-bean-greens stew	Tilapia fish cakes	Seed bars
Fri	Sopa paraguaya	Duck locro	Baked tilapia + sweet-potato mash	Chipa
Sat	Mushroom-egg tortilla	Duck or goat curry	Tilapia fish cakes	Seed bars
Sun	Mushroom-egg tortilla	Goat-bean-greens stew	Baked tilapia + sweet-potato mash	Chipa

Every day fresh: milk, yogurt or kefir ~2.2 L; eggs ~2–3; the weekly basket (salad greens, herbs, scallion, moringa, chillies, avocado, citrus, guava, banana). The man eats about half the food; he adds extra milk, seeds and honey if energy runs low. In week B swap sopa with tortilla and chipa with bars.

Stored versus fresh (per day and per cycle)

Route	What	Amount	kcal/day	Protein g/day
★ Frozen cooked meals	All 9 meals from section 8, 4 cook weeks	~5.5 kg/day; ~504 kg and ~630 L per cycle	~7,150 (at ~1.3 kcal/g)	~412
★ Fresh, daily and weekly	Milk 2.2 L/day, ~900 eggs/yr, fruit 208 kg/yr, avocado 64 kg/yr, salad and herbs ~67 kg/yr	Milk ~15.7 L, eggs ~17, fruit ~4.0 kg, avocado ~1.2 kg, salad ~1.3 kg per week	~2,382	~107
Total eaten			**~9,532** (+8% vs 8,800)	**~519** (+17% vs 445)

Frozen protein is the need-list protein minus the fresh part, scaled by 1.1 to match the larger frozen mass. This replaces my earlier rough statement that protein is ~5% above target: the item-by-item sum puts it nearer +17%, because meat, fish and egg margins are large. Both numbers carry the table’s ~20% error.

Cook-week inputs per cycle (3 people)

Job	Amount per cycle	Where it comes from
Cassava / sweet potato	~50 kg (one block dug) / ~68 kg	Cassava stays in the ground; sweet potato from the cured store
Maize / cowpea / pumpkin	~29 kg / ~28 kg / ~32 kg	Dry store and cured store
Tilapia seined	~96 kg whole (~33 kg fillet)	One of five ponds per cook week
Meat	~28 ducks, plus ~4 kids at the Feb and Aug weeks (8 a year)	Barn slaughter room; raw goat from Feb and Aug is frozen for May and Nov
Greens / tomato-pepper sauce	~27 kg / ~42 kg	Tomato and pepper frozen whole earlier
Eggs into meals	~425	Use egg-light recipes
Mushrooms	~15 kg	Fresh and dried

Balanced seven-day cook week (crew of 3)

Day	Work	Output
1	Slaughter and butcher ducks and kids in the work barn; seine one pond and fillet tilapia; dig cassava, cut greens, thaw tomato-pepper, start bone broth, soak beans, waste to BSF	Prepped meat, ~33 kg fillet, broth on
2	Strain broth; stew and locro bases, cook, cool, bag flat	~72 portions (~90 kg)
3	Curries and tomato-pepper sauce (~42 kg); chipa dough and bake; cool and bag	~79 portions (~99 kg)

Day	Work	Output
4	Sopa paraguaya bake; tortilla squares; cool and bag	~100 portions (~125 kg)
5	Baked tilapia in sauce; fish cakes; cool and bag	~100 portions (~125 kg)
6	Seed, honey and banana bars; vacuum seal, label, load freezers oldest at front, log inventory	~50 portions (~62 kg)
7	Clean kitchen and processing room, waste to BSF and compost, bones to bone meal; rest	Inventory sheet filed

Workload warning: ~500 kg of food in 6 days is heavy for 3 people. Use 2–3 helpers or split the cook week in two halves, and use the veranda as overflow workspace. Freezer, chill-room and power alarms protect the week's output.

Storage

Store	Size	Holds
★ Freezers	5 × 300 L = 1,500 L	One cycle needs ~630 L free; the rest is backup plus whole tomato, pepper, fruit, bones and offal for dogs. Cooked meals keep best 3–4 months, so the 13-week cycle leaves little margin: seal flat, date everything.
Dry goods (barn)	14.4 m²	Maize, cowpea, peanuts, sunflower seed, chia, honey, dried mushrooms: ~300 kg a year in sealed jars and bins
★ Duck grain	Grain store 16 m² + 2 steel silos × 3 m³	~1,800 kg a year of duck maize (4,500 kg of capacity covers a poor double sowing and a drought reserve)
★ Root cellar / cool store	~8 m²	Cured sweet potato (3–4 months) and pumpkin (3–6 months), up to ~270 kg and ~130 kg
★ Solar dehydrator	~4 m²	Mushrooms, fruit, herbs, chilli
In the ground		Cassava, dug one block per cook week
Fridge	Daily fridge-freezer + chilled pantry	Eggs (4–5 weeks), cheese, yogurt, kefir. Rule: harvest no more than fridge life, or cook and freeze.

Countdown around each cook week

When	Do
18 weeks before	Set Muscovy eggs under mothers (35 days to hatch, then 12–14 weeks growing)
4 weeks before	Count freezer space (need ~630 L free of 1,500 L), clean freezers, order salt, bags and vacuum rolls, check bean and dry stores; test freezer alarms and phone notifications
1 week before	Fix helpers, sharpen knives, check power, battery state, generator and ATS, confirm which pond to seine, plan which kids go (Feb and Aug)
The day before	Fast the ducks overnight, thaw whole tomato and pepper, set up the mudroom-to-kitchen flow, pre-chill the chill room and confirm its temperature log
Days 1–7	Follow the timetable; log weights and kg per dish; photograph the freezer layout
After	Record yields, egg counts and labour hours; update the nutrient balance and next cycle's areas; review dashboard fault history

9. Animals and feed

Animal	Number	Housing	Range	Notes
★ Muscovy ducks	26 hens + 6 drakes; ~28 growing per hatch, 4 hatches a year (~110 table ducks)	Raised house 6 × 6 = 36 m², straw, nest boxes, predator-proof, automatic coop door with manual lock	Pasture 40 × 40 = 1,600 m² in 4 cells of 400 m² (~27 m² per bird); gated path to ponds 3–4	Natural mothers, no incubator; never mix with other duck breeds
★ Dairy goats (Anglo-Nubian cross)	7 does, 1 buck, 2 wethers, kids	Shed 48 m² raised slatted floor, buck pen 20 m², auto waterer, mineral feeder	8 paddocks of 0.5 ha (4 ha), 7 days each = 56-day rest; shade, tannin browse; solar electric gates	Buck penned apart except for breeding; ~175 m from the mudroom
★ Tilapia + traira	5 ponds × 500 m²	One deep pond, covered for winter; 5 solar aerators	Duck path along ponds 3–4	Self-breeding; traira limit overbreeding
★ Black soldier fly	~24 totes	Shaded shed ~60 m² at the barn, temperature and humidity sensors	Self-harvest ramps	Key protein for ducks and fish
★ Bees	8–10 hives + 2 stingless boxes	Spread over 3–4 sites (orchard edge, grain edge, woodlot edge)		Honey and pollination, very low labour
★ Dogs	1–2 guard dogs + 1–2 companion dogs; livestock-guardian dog later	Guard run 64 m² at the gate; dog yard 208 m²; dog door to the mudroom	Compound only; apart from ducks and goats until trained	Fed from offal, fish frames, surplus eggs plus a bought supplement

A small chicken run is not in the baseline; add one only if the first-year egg data shows a shortfall, with the Khaki flock from v2 as the fallback.

Breeding rhythm and care

- **Muscovy:** four hatching rounds a year (~28 ducklings each) timed so ducks reach 12–14 weeks at each cook week. Incubation ~35 days: set eggs ~5 weeks before each hatch. Keep 4–7 of the best young hens a year; swap drakes with a neighbour every 2–3 years.
- **Goats:** two groups (4 and 3 does). Group A bred Mar–Apr (kids Aug–Sep), group B Sep–Oct (kids Feb–Mar); at least 4 does in milk all year; dry off ~2 months before kidding. About 8 wether kids a year raised to ~6 months for the Feb and Aug cook weeks. Swap bucks every 2–3 years.
- **Goat care (daily, non-negotiable):** milk; heat-stress check in summer; shade and level-sensor water; weekly FAMACHA check in summer; rotate paddocks; hooves every 3 months; mineral block and salt always. With 56 days of rest per paddock, parasite load should fall; verify with FAMACHA.
- **Ducks and fish:** collect eggs by hand every 1–2 days; night lock-up mandatory; move runs weekly; weekly water and feeder check; cover or move fish if water falls below 20 °C; BSF fed twice a week, larvae collected every 2 days.
- **Predators and health:** dogs, foxes, birds of prey; solid fence between goat and duck areas; quarantine newcomers 2–3 weeks; vet kit.

Feed grown on the property

Feed	Need	Supply	Status
★ Goat forage and browse (napier, moringa, pigeon pea, mulberry, sweet-potato vines, cassava leaf)	~6,600 kg DM/yr (8 adults + kids, ~16 kg/day, +10%)	4 ha at 0.5 kg DM/m²/yr = ~20,000 kg (v5.2 assumed 1.2 kg/m²); plus crop residues and a hay reserve	Large margin; a drought hay reserve of ~1,650 kg DM (~110 bales) is stored in the hay shed
★ Duck grain + protein	~1,800 kg/yr (grain equivalent)	0.5 ha maize × 0.7 = ~3,500 kg (single sowing ~1,750 kg) + BSF + duckweed + orchard fallen fruit + scraps	Covered even if the BSF trial fails; BSF and duckweed become bonus
★ Fish feed	Natural food + supplement	Duckweed 250 m² (~10,000 kg fresh), BSF, duck manure, second-grade roots	Adequate
Calcium (hens, goats)	Small	Crushed eggshell + bone meal	Closed loop

Everything the animals eat is grown on the property; the only bought items are mineral block, salt, vet kit and the dog supplement. Goat and duck manure, BSF frass and pond water go to compost and beds; bed runoff never goes to ponds.

10. Water

Peak month (driest, Aug–Sep). All values are planning estimates; replace them with a monthly balance before earthworks.

Use	Driver	Peak m³/day
★ House + processing	3 people ~200 L each incl. laundry, dogs, cook-week washing	0.8
★ Lap pool	112.5 m² evaporation ~5 mm/day, cover when idle	0.6
★ Livestock	Goats, ducks, BSF, dogs, washing	0.5
★ Fish ponds (net top-up)	2,500 m² × ~2 mm/day net of dry-season rain (pan evaporation minus rainfall; seepage extra)	5.0
★ Beds and shelters drip	~3,600 m² planted, ~half growing at once, ~2.5 L/m²/day	5.0
★ Orchards drip	~170 trees and mats × ~35 L (young trees; falls as mulch and roots mature)	6.0
★ Grain, pasture	Rain-fed; no irrigation planned	0
Total peak		**~18**

Supply / storage	Quantity	Role
★ Rain harvest	~700 m² of roof (house, veranda, annex, barn, hay shed) × ~1,500 mm × ~85% = ~900 m³/yr (rainfall assumed; verify locally)	Drinking, house, pool top-up, nursery

Supply / storage	Quantity	Role
★ Rain tanks	7 × 10 m³ (4 at the house, 3 at the barn) + 10 m³ header tank = 80 m³	~4–5 days of total peak demand, or ~40 days of house + pool + livestock (1.9 m³/day)
★ Dam	~7,470 m² surface, ~9,000 m³	Irrigation, pond top-up, fire reserve; peak draw ~16 m³/day plus ~3 mm/day net evaporation over 120 dry days is roughly 4,600 m³, about half the volume if it fills. Needs a catchment and runoff check.
★ Well	3 × 3 m pump house, ~100 m from goats, solar-direct pump	Hard gate: sustained test of at least 10 m³/day in the driest month (the dam and tanks cover the rest). Spare hand pump.
★ Header tank (gravity)	10 m³ on the high west ground, fed by a solar pump from the dam or well	Gravity drip for beds and orchards; several days of buffer
★ Greywater	~300 L/day to orchard mulch basins via grease trap	Never to pool, ponds or well
★ Swales + keyline	Set after the survey (indicatively ~1,500 m)	Rain capture, soil water, slower runoff

Rules kept from v5.2: leak detection, tank-level monitoring and alarms, first-flush diverters with automatic tank switching, float switches; bed runoff, pool water and backwash never reach ponds, beds or the well; water rights and the well test come before building.

11. Power and energy

Load (daily average, estimates)	kWh/day	Bus / note
★ 5 chest freezers (1,250 kWh/yr)	3.4	Critical
★ Daily fridge-freezer	1.2	Critical
★ Starlink + router, controllers, gateway, ~12 cameras	1.9	Critical (~80 W continuous)
Animal automation (coop door, gates, feeders, waterers)	0.3	Critical
Vents, mushroom room, BSF fans	1.0	Critical
Critical subtotal	**7.8**	**3 days = 23 kWh; 4 days = 31 kWh**
Lights, TV, PC, gaming, audio	3.0	Comfort
Kitchen (induction, oven, dishwasher, vacuum sealer)	3.5	Comfort
Laundry, workshop, tools	1.5	Comfort
Fans + split AC (summer)	3.0	Comfort
★ Outdoor audio, projector, garden and pool lighting, NAS	0.8	Non-critical, shed first
★ Games pavilion, hot tub circulation, court lights	1.2	Non-critical, shed first
Gym + music room	0.7	Non-critical, shed first
Pool pump + filtration	3.0	Solar-direct, shed first
Pond aerators (5 × 50 W × 12 h)	3.0	Solar-direct
Well + irrigation pumps	1.5	Solar-direct plus tank buffer
Typical daily total	**~29**	Plus the chill room at cook week (peak, on battery)

System	Size	Notes
★ Solar array	18 kWp: 45 × 400 W (~90 m² of panels), ground-mounted beside the compound	About 4 kWh per kWp per day on average (~70 kWh/day); a cloudy winter week may give ~20–25 kWh/day, so the generator fills gaps
★ Battery	30 kWh usable	Critical loads ~3.8 days; the whole house ~1 day
★ Inverter	12 kW hybrid	Peak loads: induction hob + oven, pool pump, well pump
★ Generator + ATS	10 kVA diesel or dual fuel, fuel for ~3 days, hand-start fallback	Last backup; fuel stored in the barn, not the house
★ Firewood and charcoal	Woodlot 4 ha	Wood oven, braai, sauna (if wood-fired), biochar

Shedding order: freezers and alarms, controllers and network, water pumps and tank valves, animal doors/gates/feeders, vents and mushroom gear, chill room (except at cook week), comfort loads, then gym, pool and aerators. Measure real loads once installed and re-size the array.

12. Automation, monitoring and security

Principle kept from v5.2: automation is a tool for reliability and labour reduction, not a goal. Sensors, simple mechanical systems and a spare-parts kit beat complex robotics, and every function has a manual override. Priority order: water, power, animal basics, environment, monitoring.

Area	Devices and methods	Fallback
Water + irrigation	Solar pumps with float switches, header tank, soil-moisture and EC/pH sensors driving zoned drip, weather station, leak detection, tank-level alarms, dam-level sensor	Manual valves on every branch, gravity tank buffer, spare pump and valves
Feed + animals	Gravity or timed duck feeders, steel silos, auto waterers with level sensors, mineral feeders, solar electric gates for goat paddocks and duck cells, automatic coop door, BSF totes with climate sensors, cameras at duck house, goat shed, BSF	Gravity hoppers, manual gate latches, coop-door manual lock, hand-feeding routine on the mudroom wall
Power	Inverter wall, 30 kWh battery, ATS + generator, freezer probes, battery state-of-charge alerts, priority load shedding	Generator started by hand; freezers kept shut
Climate + processing	Shelter and shade-house vents, mushroom-room sensors, chill-room alarm, one-way cook-week flow	Manual vent props, spare fans and misters

Area	Devices and methods	Fallback
★ Network across 45 ha	Fibre or Starlink at the house; LoRaWAN gateway on a mast for far nodes (dam, well, grain, goat paddocks, fence energisers, perimeter gate); Node-RED, MQTT, Grafana dashboard with phone alerts	Local alarms (sirens, lights) if the network drops; paper checklist
★ Security	9 cameras in the compound, farm-gate camera and intercom, gate alerts, perimeter-gate sensor, dog-yard camera, pool check-in alert, motion lighting, alarm panel with siren and panic buttons	Dogs, wall, local sirens
Explicitly excluded	Long chain-disk conveyors, full robotic feeders, complex AI pest systems, automated egg collection	

Spare-parts kit and workbench (mandatory, ranked above any extra automation): a spare of each pump type, valves, filters, sensors, irrigation fittings, belts, fuses, gate and door actuators, cable and connectors, fence energiser parts, tractor service items. Labelled bins in the barn workshop; inventory list on the dashboard; restocked each quarter.

13. Labour, fire and resilience

- **Target:** average at most 5–6 working hours a day for the main operator outside the four cook weeks (unverified; log actual hours in years 1–2 and re-set from data). With three adults the work is shared; 45 ha adds fence, track and firebreak maintenance, so keep the forest reserve and reserve land unmanaged.
- **Daily:** goat milking and checks, eggs, water, feed. **Weekly:** move duck cells and goat paddock, BSF collection, bed check, mowing with the tractor as needed. **Quarterly:** hooves, sensor calibration, spare-parts and manual-override test. **Seasonal:** planting, major harvests, track and fence repair.
- **Equipment (new):** 40–50 hp 4WD tractor with front loader, 1.6 m flail mower, 3-point rotary tiller, trailer and post-hole auger; utility vehicle (UTV); wheelbarrow set; chainsaw and splitter; weather station; walk-behind tools for beds.
- **If an operator is sick or hurt for 1–2 weeks:** written daily checklist on the mudroom wall; a trusted neighbour for goat milking; automation covers water, feeders, doors and alarms; frozen stock is the food buffer; dashboard shared with the neighbour; no slaughter or cook week within a week of travel; dry off does early if an absence is planned.
- **Fire plan (new):** 8 m firebreaks around buildings, woodlot and along the dry-side boundary; fire reserve and pump connection at the dam; extinguishers at every building door; the compound’s 4 m belt uses low-resin native and fruit species; burn permits through the local extension office.
- **Safety:** first-aid and vet kits, a written emergency plan with the nearest clinic and vet, pool safety kit, fuel stored away from the house.

14. Complete inventory with sizes and quantities

Everything on the property in one list. ★ marks new or changed items in v6.1; all sizes are planning figures.

A. Land, access and water

Item	Size / quantity	Notes
★ Property	45 ha (assumed 600 × 750 m)	Perimeter ~2.7 km
★ Farm entrance gate	5 m gate, camera, intercom, post box, 60 m² parking	South boundary
★ Boundary fence	~2.7 km simple wire fence + gates	Not the security line
★ Perimeter track	~2.7 km × 4 m	Also the outer firebreak
★ Internal tracks	~1.2 km × 4 m	All-weather gravel to barn, ponds, dam, goats, grain, orchard
★ Windbreaks	~1.2 km, 3 rows, south and east edges	Plus fodder and nitrogen-fixing alleys ~2 km
★ Native forest + wildlife reserve	~27 ha incl. spare, with wildlife corridors	Legal minimum to confirm
★ Dam	~7,470 m², ~9,000 m³, spillway, off-take, fire pump point	MADES approval
★ Wildlife wetland	~4,970 m²	Overflow from the dam
★ Swales + keyline contours	~1,500 m (set by survey)	
Rain tanks	7 × 10 m³ + 10 m³ header = 80 m³	
★ Header tank (gravity)	10 m³	High west ground
Well + pump house	3 × 3 m (9 m²), hand pump	~100 m from animals and humanure

B. Compound (5,040 m²)

Item	Size / quantity	Notes
★ Wall	2.4 m high, 284 m	Solid, with motion lighting
★ Gates	5 m sliding (motorised), 1 m pedestrian with intercom, 3 m service gate	
★ Court	198 m²	Parking and turning
★ Guard run + kennel	64 m² + 2 × 2.5 m kennel	At the gate
★ Dog yard	208 m², dig pit 4 m²	Dog door, wash, shade, tap
★ Service yard	220 m²	Harvest, fuel, trades
★ Tree belt	4 m wide, 1,072 m²	Native and fruit/flower species
★ Compound orchard	440 m², 16 trees	Greywater basins
★ Showpiece beds + lawn	~830 m²	Test beds, soil journal, sensors
★ Herb + edible flower garden	63 m²	
★ Fire pit + seating	~50 m²	
★ Hammock grove	40 m²	
★ Quiet nook	1 bench	
★ Outdoor solar rinse shower	1	Beside pool end
★ Wildlife pond + boxes	~30 m²; 6 bat and bird boxes	
★ Garden shed	15 m²	
★ Sauna + cold plunge (optional)	15 m²	
★ Lighting	Warm low-level: veranda, paths, pool, fire pit	Motion lights at gates and corners
★ Optional catio	~6 m²	Only if cats

C. House (280 m²) and veranda

Item	Size / quantity	Notes
Potting + seed room	20.0 m²	
Living + dining + TV / sport wall	40.0 m²	
★ Media + gaming lounge	20.9 m²	Darkenable
★ Kitchen	22.8 m²	
Pantry	11.4 m²	
★ Freezer room	11.4 m², 5 × 300 L	
Processing + storage	17.1 m²	
Mudroom, laundry, inverter wall, dog wash	22.8 m²	
Main bath	12.0 m²	
Master + walk-in + ensuite	22.0 + 6.0 + 6.0 m²	
Bedroom 2	17.0 m²	
★ Bedroom 3 / study	17.0 m²	

Item	Size / quantity	Notes
Corridor	33.6 m²	
★ Veranda	98 m² (56 roofed + 42 pergola)	Outdoor kitchen bay with wood oven (~2 m²)
★ House systems	Solar hot water 2 × 200 L; screens on all openings; ceiling fans; UV + carbon water filter; surge and lightning protection; daily fridge-freezer; Starlink + mobile backup	

D. Annex and pool

Item	Size / quantity	Notes
★ Gym	39 m²	
★ Music room	27 m²	
Covered patio	27 m²	
Lap pool	25 × 4.5 m, ~150 m³; pool + decks 210 m²	Cover, fence, alarm, dog ramp
★ Plant room	20 m²	Filter, pump, salt cell

E. Food production

Item	Size / quantity	Notes
★ Annual crops (planted)	3,365 m²	See section 7 for each crop
★ Rain shelters	6 × 45 m² = 270 m²	
★ Shade house + seedling tunnel	~100 m²	
★ Perennial staples	~800 m²	Taro, perennial spinach, pigeon pea, moringa
Mushroom room	25 m²	
★ Main orchard	1.5 ha: ~80 trees, 40 banana mats, 40 yerba mate	
★ Grain fields	1.0 ha (0.5 ha duck maize + 0.5 ha rotation)	
★ Reserve	2.0 ha cover crop	
★ Duckweed	250 m²	
★ Vermicompost bay + biochar pad	~6 m² + ~30 m²	
★ Woodlot	4 ha	

F. Animals

Item	Size / quantity	Notes
★ Goats	7 does, 1 buck, 2 wethers + kids; shed 48 m²; buck pen 20 m²; 8 paddocks of 0.5 ha; ~1,600 m of goat fence	
★ Muscovy	26 hens + 6 drakes; house 36 m²; pasture 1,600 m² in 4 cells	
★ Ponds	5 × 500 m²; 5 solar aerators	
★ BSF	~24 totes; shed 60 m²	
★ Bees	8–10 hives + 2 stingless	
Dogs	1–2 guard + 1–2 companion	

G. Leisure and entertainment

Item	Size / quantity	Notes
★ Outdoor sound	6 weatherproof speakers + amplifier	
★ Garden cinema	3 m pull-down screen, projector, 4 loungers	
★ Pool-side comfort	4 loungers, 20 m² shade sail, pool lighting	
★ Games and reading nook, coffee corner	1 table, 3 m shelf, daybed, drinks fridge	
★ NAS media server	20 TB	
★ Games pavilion + bar	56 m²: pool table, table tennis, darts	
★ Hot tub	3 × 3 m, seats 4	
★ Multi-sport court	10 × 20 m = 200 m², solar LED lights	
★ Recording corner	1 desk, interface, monitors	
★ Stargazing deck + telescope pad	16 m² + 0.6 × 0.6 m pier	
★ Outdoor cooking table	6 m²	
★ Run and cycle loop	2.7 km, 2 water points, 3 bikes	
★ Forest trail	~3 km, 2 benches, 4 wildlife cameras	
★ Bird hide	5 m²	
★ Dam shelter + fishing deck	30 m² + 12 m²	

Item	Size / quantity	Notes
★ Kayaks, camp pad (optional)	2 kayaks; 8 × 8 m pad	
★ Hammocks, bird feeders, nest and bat boxes	4 hammocks, 6 feeders, 6 boxes	

H. Barn, utilities and equipment

Item	Size / quantity	Notes
Work barn	125 m²: slaughter/wet room 40, workshop 16, grain 16, dry goods 14.4, spare parts 14.4, chill room 12, wash/PPE 12	
★ Hay + machinery shed	~150 m²	Hay ~110 bales reserve
★ Grain silos	2 × 3 m³	
★ Zone C	1,600 m²: humanure, compost, vermicompost, firewood, biochar	
★ Solar field	18 kWp, 45 × 400 W, ~600 m² footprint with battery and generator sheds	30 kWh battery, 12 kW inverter, 10 kVA generator + ATS
★ Solar dehydrator + root cellar	~4 m² + ~8 m²	
★ Vehicles and tools	Tractor 40–50 hp + implements, UTV, trailer, wheelbarrows, chainsaw, splitter	
★ Weather station + LoRa gateway	1 + 1	

15. Build order, trials, rules, risks and legal checklist

Phase	Build	Cost
0–6 months	Boundary and topographic survey; water rights and sustained well test; dam feasibility; INFONA and MADES checks; soil tests; sun-path and wind check; access road; confirm local supply of sensors, valves, LoRa and Node-RED-compatible hardware	TBD
0–9 months	Rain tanks, compound wall and gates, farm entrance and boundary fence, tracks, first spare-parts kit, temporary power	TBD
6–18 months	House core (222 m² + veranda), work barn, well and pump, solar field and battery, food block (staples, shelters, shade house), compound orchard, herb garden and showpiece beds, dog yard and guard run, drip and soil sensors, dashboard and LoRa gateway; plant the main orchard and woodlot in the first wet season	TBD
12–24 months	Dam (dry-season works), BSF, Muscovy and pasture, grain fields, tractor, auto doors, feeders and gates; annex (gym and music room)	TBD
18–30 months	Goats and 8 paddocks, ponds (in two stages), bees, mushroom room, five freezers, power system finalised; west wing (potting, media lounge), bedroom 3	TBD
24–36 months	Pool and plant room, sauna and plunge (optional), first complete cook weeks, nutrient and water re-runs from real data	TBD

Proposal: hold a contingency reserve of 15–20% of build cost plus one year of running costs. Big new capital items versus v5.2 are the dam, tracks and fence, solar field, tractor and the larger house and annex. No prices were supplied, so none are invented here.

Trials scoreboard (years 1–2)

Trial	Pass line	If it fails
Well	≥10 m³/day in the driest month	More tank storage or dam, or stop
★ Dam	Fills in the first wet season and holds through the dry season	Smaller draw, more tanks, repair or re-site
Double sowing	Both sowings reach 0.60 / 0.37 / 0.078 kg/m² a year (maize / cowpea / sunflower)	Keep the 1.5× planting and use the reserve
Tomato + pepper beds; greens and salad	≥3.2 kg/m²; ≥1.8 / ≥3 kg/m²	Enlarge beds or add shelters
Chia	≥0.1 kg/m²	Switch to perilla, then algae oil
Mushrooms	~1 kg per bag-cycle	More bags or buy spawn
Duck grain	Maize ≥0.35 kg/m² even at a single sowing	Add maize area; BSF output decides the rest
Eggs	≥2,600 a year	Khaki fallback or small chicken run
Milk	≥1,200 L a year, ≥4 does always in milk	Add a doe
★ Fish	≥160 kg fillet a year	More aeration, feed, or a sixth pond
★ Power autonomy	3 days of critical loads on battery, tested before the first cook week	More battery or fewer critical loads
★ Pool	Water loss ≤0.8 m³/day in summer	Cover use, leak check
Labour	≤5–6 h/day logged for the main operator	Cut scope or hire help
Sensors, LoRa and dashboard	Alerts reach the phone; no unexplained dropout over a season; manual overrides tested each quarter	Simplify, replace sensors, add local alarms

Hard rules

#	Rule
1	Every nutrient has at least two food sources where possible.
2	Re-run the nutrient balance whenever species or counts change (especially after Muscovy egg data).
3	Keep ~30% surplus on every crop and animal food (eggs: see decision).
4	Harvest no more than fridge life, or cook and freeze immediately.
5	Trial any new species on 20 m² before scaling (chia, tomato/pepper in soil beds, mushrooms, BSF output).
6	Never send bed runoff into the fish ponds.
7	Cover or move fish if water falls below 20 °C.
8	Check goats daily for heat stress in summer.
9	Increase soil organic matter every year.
10	Legal checks before building, slaughtering, selling, clearing or digging a dam.
11	Confirm double sowing of maize, cowpea and sunflower, or keep the 1.5× planting.
12	Trial BSF output before finalising the maize area.
13	Every automated function keeps a manual override.
14	Keep the spare-parts kit complete; it outranks adding automation.
15	Pool water, backwash and chlorine never reach ponds, beds or the well.
16	★ The compound wall is the security line; the 45 ha boundary only gets a simple fence.
17	★ The well stays about 100 m from animals, humanure and compost, and at least 50 m from ponds.
18	★ Confirm forest-reserve and dam approvals before earthworks.
19	★ Plant at 1.5× the minimum area until yields are proven.
20	★ Re-check the veranda sun geometry against the surveyed orientation.

Open risks

Risk	Mitigation
Double sowing unreliable	1.5× planting; reserve land
Egg shortfall (Muscovy)	Egg-light recipes; Khaki fallback; re-run balance
Soil-bed yields guessed; chia untested; ALA and vitamin D gaps	20 m² trials; perilla; algae oil; bloodwork
Sandy soil years 1–3	Compost, biochar, cover crops, lime + rock phosphate
MADES may restrict tilapia; growth stops below ~20 °C	Check rules first; deep covered pond; harvest by weight
★ Water: 18 m³/day peak, pond and dam evaporation, dam approval	Well test, monthly balance, dam hydrology check, tanks as buffer
★ Forest-reserve and land-use rules unknown for this plot	INFONA and MADES check before any clearing; re-place blocks around existing forest
★ Wind assumption (goats north-north-west of compound)	Survey; swap goat block with grain fields if the wind is from the north or north-east
★ Long daily walk to the goat hub (~175 m)	Utility vehicle; consider moving the hub if the well can move
★ Leisure items cost power, upkeep and add fire risk	Non-critical bus, gravel fire aprons, phase priority 3 after year 2
★ Workload from 45 ha and a larger house	Phase the build; keep reserve and forest unmanaged; log hours; hire help
★ Pool and sauna cost, upkeep and safety with dogs	Cover, fence, alarm, solar-direct pump, non-critical bus; cut sauna first
Goat heat stress and parasites	Shade, 56-day paddock rest, FAMACHA checks
Mushroom contamination	Clean room, pasteurised substrate, small batches
Freezer or power failure	30 kWh battery, generator + ATS, freezer alarms, load shedding
Sensor, pump or network failure	Spare parts, manual override, local alarms, paper checklist
Operator illness or injury	Checklist, neighbour, automation, frozen stock

Legal checklist

- Municipalidad de Villarrica: zoning, building permits.
- MADES: water rights, well, dam and water-body setbacks, fish species.
- INFONA: forest reserve, clearing and land-use permits.
- SENACSA: slaughter (work barn), dairy, animal health.
- Local extension office: bees, composting, burn permits, food safety, vaccines.
- Cadastre, title and boundary survey before fencing and tracks.

Closing notes

- All yields remain untested planning estimates. The first two years are for proving yields, refining labour, building soil and shaking down the automation; only after that data exists should the system be considered truly self-sufficient.
- Philosophy: two layers. The private compound is for daily enjoyment (nature, training, swimming, screens, music, cooking, rest, pets); the productive land is for reliability, nutrition, low labour and closed loops. Life-critical systems win any conflict.
- The site plan, compound and house are concepts built on assumed wind, slope and plot shape. Confirm them on the survey, then redraw scaled construction drawings.